CURRICULUM VITAE (February 25, 2026)
ALEXANDER G. KVASHNIN

Personal data
Born on 05.08.1989 in Krasnoyarsk, Russia
Married, two children
Languages: Russian (native), English (fluent), Chinese (HSK3)

ORCID: http://orcid.org/0000-0002-0718-6691
Scopus Author ID: 6603118309
ResearcherID: A-1825-2014
Google Scholar page
Group page

EDUCATION

2021	Doctor Habilitatus in condensed matter physics (Doctor of Physical & Mathematical Sciences, Russian highest degree), National University of Science and Technology "MISiS" (thesis)
2016	Ph.D. in condensed matter physics (Candidate of Physical & Mathematical Sciences) Moscow Institute of Physics and Technology (thesis)
2012	M.Sc. in Applied Mathematics and Physics, Moscow Institute of Physics and Technology
2010	B.Sc. in physics, Siberian Federal University

EMPLOYMENT

2023-present	Full Professor, Skolkovo Institute of Science and Technology, Moscow
2021-2023	Assistant Professor, Skolkovo Institute of Science and Technology, Moscow
2021-present	Professor, National University of Science and Technology MISiS, Moscow
2017-2021	Senior Researcher, Skolkovo Institute of Science and Technology, Moscow
2015-present	Research Scientist, Emanuel Institute of Biochemical Physics RAS, Moscow
2016-2017	Research Scientist, Skolkovo Institute of Science and Technology, Moscow
2015-2016	Junior Researcher, Skolkovo Institute of Science and Technology, Moscow
2013-2015	Junior Researcher, Technological Institute for Superhard and Novel Carbon Materials (TISNCM), Troitsk, Moscow
2012-2020	Administrator of HPC cluster, Technological Institute for Superhard and Novel Carbon Materials (TISNCM), Troitsk, Moscow
2010-2012	Research Intern, Technological Institute for Superhard and Novel Carbon Materials (TISNCM), Troitsk, Moscow

Ph.D. Advisor: Pavel B. Sorokin (NUST MISIS)
Postdoctoral Advisor: Artem R. Oganov (Skoltech)

CITATION SUMMARY: h-index: 31 (Scopus), 36 (Google Scholar); Citations: 6200 (Scopus), 8500 (Google Scholar)

VISITING APPOINTMENTS:
Visiting International Scientist, Rice University, Houston TX, USA, 2011, 2013;
Visiting PhD student, University of Namur, Namur, Belgium, 2015;
Visiting Professor, Northwestern Polytechnical University, Xi'an, China, 2018.

HONORS AND AWARDS

2024	Sber Science Prize in the nomination “AI in Science. Physical World”
2023	Ranked in the top 2% of the world's most cited researchers by single-year impact, as compiled by Elsevier.
2023	Excellence Recognition Award for results in research, teaching and student success, economic impact and service to Skoltech and wider community
2022	Excellence Recognition Award for results in research, teaching, impact, and service to Skoltech and wider community
2020	Prize of the Moscow Government for Young Scientists in Science and Innovations
2020	Award of Academia Europaea for young scientists in physics
2013-2015	Scholarship of President of Russia for young scientists and PhD students
2010-2012	Scholarship of the “Dynasty” foundation
2010-2011	L.V. Kirensky’s scholarship for achievements in the field of Mathematical and Physical Sciences
2010	Scholarship for achievements in the field of Mathematical and Physical Sciences
2009	President Scholarship for students
2009	Award of the head of the Krasnoyarsk city for young talent for excellence in scientific and educational activities

PROFESSIONAL ASSOCIATIONS

2025-present	Member of Expert Council of the Krasnoyarsk Science Foundation
2024-present	Expert of the Scientific and Technological Council of the Russian Science Foundation
2023-present	Member of Dissertation Council of RUDN University ПДС 2022.013
2022-present	Member of Research and Innovation Committee of Skoltech
2022-present	Member of Advisory Panel in Energy of Skoltech
2022-present	Member of Dissertation Council of ITMO University 12.22.00
2022-present	External expert of the Council for the priority area of scientific and technological development of the Russian Federation "Transition to digital, intelligent manufacturing technologies, robotic systems, new materials and design methods, creation of systems for big data processing, machine learning and artificial intelligence"
2018-present	Expert of the Russian Science Foundation

ASSIGNED REVIEWER

The Innovation, Ceramics International, Advanced Materials, Nano Letters, The Journal of Chemical Physics, Computational Materials Science, 2D Materials, Journal of Applied Physics, The Journal of Physical Chemistry Letters, Carbon, Physica E, Solid State Communications, Materials, Nanomaterials, Membranes, ACS Omega, Advanced Science

GRANTS AND PROJECTS

2024–2026	Russian Science Foundation project № 24-23-00125, "Computational design for catalysts based on higher molybdenum and chromium borides for energy transition problems" **(leader)**
2023	Megagrant of the Ministry of Science and Higher Education of the Russian Federation № 075-15-2021-579, "Computational synthesis of chemical compounds" **(leader)**
2020-2025	Program of Introducing Talents of Discipline to Universities of the Ministry of Education of China "Integrated Computational Materials Engineering in Metals and Alloys project" (**key participant**)
2021–2023	Grant of the President of the Russian Federation for young scientists МК-3120.2021.1.2 **(key participant)**
2021–2022	Russian Foundation of Basic Research project № 20-53-05009, "Search and investigation for novel two-dimensional materials for biochemical sensing applications" **(key participant)**
2020-2023	Russian Science Foundation № 20-12-00097, "Quantum mechanical modeling and experimental studies of new superhard and high-strength states of carbon" (**participant**)
2020–2021	Grant of Ministry of Science and Education of Russian Federation for support of leading scientific schools of the Russian Federation № НШ-2711.2020.2 (**participant**)
2019–2020	Russian Foundation of Basic Research project № 19-03-00100, "Study of superconductivity with electron-phonon coupling mechanism in hydrides, borides and carbides of transition metals stabilized by pressure" **(leader)**
2018–2020	UMNIK program, Light elements: superconducting ternary systems with light elements **(leader)**
2019–2022	Russian Science Foundation № 19-72-30043, "Computational materials design laboratory for new materials" (**key participant**)
2017–2019	Industry project with LLC "Gazpromneft' NTC", "Design new superhard materials for the cutters of the drill bit, and their subsequent synthesis and testing" (**key participant**)
2017–2019	Russian Science Foundation project № 17-73-20038, "Computational design for new materials with optimal hardness and fracture toughness" **(leader)**
2016–2018	Russian Science Foundation project № 16-13-10459, "New methods for search for materials with optimal properties" (**participant**)
2014–2016	Russian Science Foundation project № 14-12-01217, "Simulations of structure and properties of new multilayered nanomaterials based on TMDs and BN-graphene layered structures" (**participant**)
2012–2015	Research project of Federal Target program № 14.B37.21.1645. Investigation of fabrication ways and properties of single-crystal diamond film with nanometer thickness (**participant**)
2012–2015	Russian Foundation of Basic Research program № 12-02-31261. Investigation of features of electronic, elastic, and mechanical properties of materials based on diamond clusters with nanometer size (**participant**)

TEACHING

2024- present	Lecturer of the course MA060589 "Thermodynamics of Materials" for MSc and PhD students in Materials Science program in Skoltech, **6 ECTS**
2023- present	Lecturer of the course MA060573 "Computational Methods in Atomistic Simulations" for MSc and PhD students in Materials Science program in Skoltech, **6 ECTS**
2023- present	Co-lecturer of the course DA060341 "Advanced Materials Modeling" for MSc and PhD students in Materials Science program in Skoltech, **6 ECTS**
2023- present	Lecturer of the course MA030565 "Introduction to Computational Mechanics in Energy Transition" for MSc students in Advanced Computer Mechanics program in Skoltech, **6 ECTS**
2022- present	Co-lecturer of the course MA060008 "Computational Chemistry and Materials Modeling" for MSc and PhD students in Materials Science program in Skoltech, **6 ECTS**
2020- present	Lecturer of the course "Introduction to the Quantum Theory of Condensed Matter" for 3rd year BSc students in Materials Science program in NUST MISiS, **3 ECTS**
Nov.-Dec. 2022	Lecturer of the course "Programming of chemical tasks" for MSc students in ITMO University, **3 ECTS**
Mar. 2021	Lecturer of the on-line course "Simulation of materials properties" in Physics Department of Northeastern University, China
2017-2019	Co-lecturer of the course MA060075 "Structure and Properties of Materials" for PhD students in Materials Science program in Skoltech, **6 ECTS**

STUDENT SUPERVISIONS

Name	Affiliation	Years
Faridun Jalolov	MSc Skoltech	2020 – 2022
	PhD, Skoltech	2022 –
Aleksandra D. Radina	MSc MISIS	2021 – 2023
	PhD, Skoltech	2023 –
Olga Pushkova	BSc MISIS	2022 – 2024
	MSc Skoltech	2024 –
Daniil Fedorov	BSc MISIS	2022 – 2024
	MSc MISIS	2024 –
Artem Butyaev	BSc MISIS	2024 –2025
	MSc MISIS	2025 –
Mikhail Lukanov	PhD, Skoltech	2024 –
Khasan Akhmadiev	PhD, Skoltech	2024 –
Anton Konkin	BSc MISIS	2025 –
Ivan Nikitin	BSc MISIS	2025 –
Grigory Bychkov	MSc Skoltech	2025 –
Daniil Alekseev	MSc Skoltech	2025 –
Daria Fedotova	MSc Skoltech	2025 –

FORMER STUDENTS

D.V. Semenok (PhD, 2018-2022, now HPSTAR, Beijing, China), N.A. Matsokin (MSc, 2021-2023, now KIT), D.A. Bystrov (BSc, MISIS, 2021-2023), T.S. Pigulevskaya (BSc, MISIS, 2021-2023), I.T. Yanaev (BSc, MISIS, 2021-2023), Labib Abdelrahman Mohamad Bakr (MSc, MISIS, 2022-2023), T. Kozlan (BSc, MISIS, 2022-2024), E.E. Fatkulina (BSc, MISIS, 2022-2024), M. Trunov (BSc, MISIS, 2024-2025).

BOOKS

1. A.R. Oganov, G. Saleh, **A.G. Kvashnin** (Editors). Computational Materials Discovery. Royal Society of Chemistry. ISBN: 978-1-78262-961-0 (2018)

PATENTS

1. Yu.Z, Vasilieva, A.A. Svinukhova, A.A. Kuznetsova, F.V. Vlasov, A.Ya. Pak, **A.G. Kvashnin**, R.D. Gerasimov, P.V. Povalyaev, Z.S. Bolatova. 2025. Method of producing powder of single-phase high-entropy diboride of composition Ti-Zr-Nb-Hf-Ta-B with hexagonal lattice, RU 2841156. Filed June 21, 2024 and issued June 3 2025
2. A.Ya. Pak, A.A. Gumovskaya, P.V. Povalyaev, **A.G. Kvashnin**. 2023. Powder production method based on single-phase high-entropy Ti-Zr-Nb-Hf-Ta-C carbide with cubic lattice, RU 2796134. Filed June 21, 2022 and issued May 17 2023
3. **A.G. Kvashnin**, I.S. Lyubutin, I.A. Troyan, D.V. Semenok, A.R. Oganov. 2021. High-temperature superconducting hydride and method for production thereof, RU 2757450. Filed September 09, 2020 and issued October 15, 2021.
4. V.V. Brazhkin, V.I. Bugakov, I.P. Zibrov, V.P. Filonenko, A.R. Oganov, **A.G. Kvashnin**, A.Ya. Zakirov. 2020, Materials based on chromium tetraboride and methods for production thereof, RU 2753339. Filed July 25, 2020 and issued August 13, 2021
5. V.V. Brazhkin, V.I. Bugakov, I.P. Zibrov, V.P. Filonenko, A.R. Oganov, **A.G. Kvashnin**, A.Ya. Zakirov, A.A. Osiptsov. 2019. Method of producing superhard material and superhard material based on tungsten pentaboride. RU 2698827. filed August 01, 2018 and issued August 30, 2019.

PAPERS IN REFEREED JOURNALS AND PREPRINTS

2026

111 A.D. Radina, A.Yu. Kurenkova, V.S. Baidyshev, P.V. Povalyaev, A.A. Saraev, R.F. Alekseev, V.A. Lomakina, E.E. Aydakov, E.Yu. Gerasimov, D.D. Mishchenko, A.Ya. Pak, E.A. Kozlova, **A.G. Kvashnin**, High-Entropy Carbide as Cocatalyst for TiO2 in Photocatalytic CO2 Reduction, Materials Today Energy, 56, 102206 (2026) (DOI: 10.1016/j.mtener.2026.102206), Q1, SJR10%

110 D.S. Nikitin, I.I. Shanenkov, A. Nassyrbayev, A.A. Sivkov, V.S. Baidyshev, Yu.A. Kvashnina, N.A. Matsokin, A.Ya. Pak, **A.G. Kvashnin**, High-entropy Ti-Zr-Nb-Hf-Ta carbide and carbonitride coatings fabricated by high-speed arc discharge plasma jet, Int. J. Refr. Met. Hard Mater., 135, 107537 (2026) (DOI: 10.1016/j.ijrmhm.2025.107537), Q1

109 R. Arabov, N. Rybin, V. Demin, M. Polovinkin, **A. Kvashnin**, L. Chernozatonskii, A. Shapeev, Tuning Thermal Conductivity and Electron-Phonon Interactions in Carbon and Boron Nitride Moir\'e Diamanes via Twist Angle Manipulation, Appl. Surf. Sci., 717, 164801 (2026) (DOI: 10.1016/j.apsusc.2025.164801), Q1

2025

108 S.G. Buga, M.Yu. Popov, D.A. Ovsyannikov, V.S. Baidyshev, A.G. Kvashnin, Admittance spectroscopy of 2–5 nm detonation diamonds in NaCl-nanodiamond mixture. Electronic properties of small nanodiamonds, Diam. Rel. Mat., 113184 (2025) (DOI: 10.1016/j.diamond.2025.113184), Q1

107 I.V. Chepkasov, **A.G. Kvashnin**, Multilayers Alkali Metal Structures a Way to High Capacity and Fast Charging Carbon-Based Metal-Ion Battery. A Review, Small, e08433 (2025) (DOI: 10.1002/smll.202508433), Q1, SJR10%

106 I.V. Chepkasov, V.S. Baidyshev, **A.G. Kvashnin**, Core-dictated tuning of the performance of amorphous and crystalline TM@Pt and HEA@Pt core-shell nanoparticles

catalysts, Mat. Today. Energy, 54, 102130 (2025) (DOI: 10.1016/j.mtener.2025.102130), Q1, SJR10%

105 Yu.A. Bayan, E.R. Beskopylny, E.U. Gerasimov, E.E. Aydakov, K.K. Volik, I.V. Pankov, I.V. Chepkasov, M.M. Lukanov, **A.G. Kvashnin**, A.A. Alekseenko, Boosting the Performance of Pt/C Catalysts via Nitrogen-Doped Carbon Support: Insights from Structural and Electrochemical Characterization, Small, e10144 (2025) (DOI: 10.1002/smll.202510144), Q1, SJR10%

104 Yu. Vassilyeva, Yu. Neklya, M. Lukanov, **A. Kvashnin**, A. Pak, Vacuum-free arc synthesis and characterization of crystalline molybdenum borides as instrumental material, Therm. Sci. Eng. Progress, 67, 104134 (2025) (DOI: 10.1016/j.tsep.2025.104134), Q1, SJR10%

103 C. Tantardini, **A.G. Kvashnin**, M. Giantomassi, M. Iliaš, B.I. Yakobson, R.J. Hemley, X. Gonze, Charge density waves and structural phase transition in the high-Tc superconducting LaH_{10} quantum crystal, Phys. Rev. B, 112, 115154 (2025) (DOI: 10.1103/c7w9-7tgy), Q1

102 A.R. Vildanova, A.E. Goldt, S.V. Porokhin, **A.G. Kvashnin**, V.S. Baidyshev, I.V. Chepkasov, F.S. Fedorov, K.A. Litvintseva, A.V. Lalov, V.A. Dmitrieva, M.M. Tepliakova, A.G. Nasibulin, Encapsulated Nickel Nanowires Inside Plasma-treated Single-Walled Carbon Nanotubes for Urea Oxidation, Small, e07040, (2025) (DOI: 10.1002/smll.202507040), Q1, SJR10%

101 D.Yu. Karpenkov, R.A. Makarin, C. Tantardini, **A.G. Kvashnin**, A.Yu. Karpenkov, E.V. Argunov, T.A. Sviridova, T.B. Shapaeva, M.V. Zheleznyi, Designing high-efficiency cryogenic regenerators: The role of microstructure and geometry in magnetocaloric cooling, J. All. Comp., 1041, 183808 (2025) (DOI: 10.1016/j.jallcom.2025.183808), Q1, SJR10%

100 I.V. Chepkasov, A.D. Radina, V.S. Baidyshev, M. Polovinkin, N. Rybin, A. Shapeev, A.A. Krikorov, A.R. Oganov, Z. Dashevsky, D.G. Kvashnin, **A.G. Kvashnin**, Tuning of Mechanical Properties of doped PbTe-based Thermoelectric Material Driven by Intrinsic Defects, J. Mat. Chem A., 13, 31170-31181 (2025), (DOI: 10.1039/D5TA00823A), Q1, SJR10%

99 C. Tantardini, M. Azizi, T. Altalhi, **A.G. Kvashnin**, A. Filippetti, C. Gatti, B.I. Yakobson, X. Gonze, Sb^{IV}, an Unusual Player in 2D Spintronic Devices, ACS Nano, 19, 29, 26562–26571 (2025) (DOI: 10.1021/acsnano.5c05027), Q1, NI, SJR10%

98 N.A. Matsokin, R.A. Eremin, A.A. Kuznetsova, I.S. Humonen, A.V. Krautsou, V.D. Lazarev, Y.Z. Vassilyeva, A.Ya. Pak, S.A. Budennyy, **A.G. Kvashnin**, A.A. Osiptsov, Discovery of chemically modified higher tungsten boride by means of hybrid GNN/DFT approach, npj Comp. Mat., 11, 163 (2025) (DOI: 10.1038/s41524-025-01628-z), Q1, SJR10%

97 I.V Chepkasov, V.S Baidyshev, **A.G Kvashnin**, Polyaromatic hydrocarbons as prospective anode materials for metal ion battery, J. En. Storage, 125, 116831 (2025) (DOI: 10.1016/j.est.2025.116831), Q1, SJR10%

96 I.V. Chepkasov, V.S. Baidyshev, A.V. Iosimovska, I.S. Zamulin, **A.G. Kvashnin**, Adsorption properties of crystalline and amorphous PdIr nanoparticles. A systematic first-principles study, J. Catalysis, 447, 116102 (2025) (DOI: 10.1016/j.jcat.2025.116102), Q1

95 D.S. Nikitin, I.I. Shanenkov, A.R. Nassyrbayev, A.A. Sivkov, V.S. Baidyshev, Yu.A. Kvashnina, N.A. Matsokin, A.Ya. Pak, **A.G. Kvashnin**, Synthesis of high-entropy Ti-Zr-Nb-Hf-Ta carbides and carbonitrides in high-speed arc discharge plasma jet, J. All. Comp., 1010, 177178 (2025) (DOI: 10.1016/j.jallcom.2024.177178), Q1, SJR10%

2024

94 F.N. Jalolov and **A.G. Kvashnin**, Physically intuitive anisotropic model of hardness, Phys. Rev. Mat., 8, 123601 (2024) (DOI: 10.1103/PhysRevMaterials.8.123601), Q1

93 V.S. Baidyshev, C. Tantardini, **A.G. Kvashnin**, Melting simulations of high-entropy carbonitrides by deep learning potentials, Sci. Rep., 14, 28678 (2024) (DOI: 10.1038/s41598-024-78377-4), Q1, SJR10%

92 C. Tantardini, H.A. Zakaryan, Z.-K. Han, T. Altalhi, S.V. Levchenko, **A.G. Kvashnin**, B.I. Yakobson, Material hardness descriptor derived by symbolic regression, J. Comp. Sci., 82, 102402 (2024) (DOI: 10.1016/j.jocs.2024.102402), Q2

91 R.S. Stepanov, A.D. Radina, C. Tantardini, **A.G. Kvashnin**, A.V. Kolobov, Chemical bonding within AIIIBVI materials under uniaxial compression, Phys. Chem. Chem. Phys., 26, 20984-20992 (2024) (DOI: 10.1039/D4CP00937A), Q2

90 A.D. Radina, V.S. Baidyshev, I.V. Chepkasov, N.A. Matsokin, T. Altalhi, B.I. Yakobson, **A.G. Kvashnin**, Theoretical study of adsorption properties and CO oxidation reaction on surfaces of higher tungsten boride, Sci. Rep., 14, 12788 (2024) (DOI: 10.1038/s41598-024-63676-7), Q1, SJR10%

89 F. Bochkanov, D. Karpenkov, V. Fomin, A. Tukmakova, **A. Kvashnin**, A. Novotelnova, O. Kutsemako, N. Kulesh, V. Kurichenko, I. Bajenova, A. Khvan, Electrical current-assisted reactive crucible melting technique: Case study of the Fe-Sn system, Materialia, 36, 102152 (2024) (DOI 10.1016/j.mtla.2024.102152), Q2.

88 A.Yu. Kurenkova, A.D. Radina, V.S. Baidyshev, P.V. Povalyaev, E.E. Aidakov, E.Yu. Gerasimov, D.D. Mishchenko, A.V. Zhurenok, A.Ya. Pak, E.A. Kozlova, **A.G. Kvashnin**, Photocatalytic H_2 generation and CO_2 reduction by WB_{5-x} cocatalyst of TiO_2 catalyst, Appl. Surf. Sci., 661, 160095 (2024) (DOI: 10.1016/j.apsusc.2024.160095), Q1

87 F.N. Jalolov, E.V. Podryabinkin, A.R. Oganov, A.V. Shapeev, **A.G. Kvashnin**, Mechanical Properties of Single and Polycrystalline Solids from Machine Learning, Adv. Theory and Sim., 2301171 (2024) (DOI: 10.1002/adts.202301171), Q1

86 I.V. Chepkasov, A.D. Radina, **A.G. Kvashnin**, Structure-driven tuning of catalytic properties of core–shell nanostructures, Nanoscale, 16, 5870-5892 (2024) (DOI: 10.1039/D3NR06194A), Q1

85 I.V. Chepkasov, **A.G. Kvashnin**, A.D. Radina, N.A. Matsokin, F.N. Jalolov, D.G. Kvashnin, A.R. Oganov, Z. Dashevsky, Origin of brittle behavior of doped PbTe-based thermoelectric materials, Appl. Phys. Lett., 124, 022104 (2024) (DOI: 10.1063/5.0185002), Q1, NI.

84 R.A. Khmelnitsky, V.P. Martovitsky, J.V. Bondareva, A.I. Kolbatova, N. Titova, G.N. Goltsman, F.S. Fedorov, A.V. Egorov, N.A. Matsokin, **A.G. Kvashnin**, D.G. Kvashnin, S.A. Evlashin, Synthesis and Characterization of Niobium Carbide Thin Films on Diamond Surface for Superconductive Application, J. Alloys and Compounds, 173266 (2024) (DOI: 10.1016/j.jallcom.2023.173266), Q1, SJR10%

83 C. Tantardini, M. Iliaš, M. Giantomassi, **A.G. Kvashnin**, V. Pershina, X. Gonze, Generating and grading 34 optimised norm-conserving Vanderbilt pseudopotentials for actinides and super-heavy elements in the PseudoDojo, Comp. Phys. Comm., 295, 109002 (2024) (DOI: 10.1016/j.cpc.2023.109002), Q1, SJR10%

2023

82 I.V. Chepkasov, I.S. Zamulin, V.S. Baidyshev, **A.G. Kvashnin**, Tuning the surface properties of AuPd nanoparticles for adsorption of O and CO, Phys. Chem. Chem. Phys., 25, 33031-33037 (2023) (DOI: 10.1039/D3CP03213B), Q1.

81 I.V. Chepkasov, V.S. Baidyshev, **A.G. Kvashnin**, Structure-driven tuning of O and CO adsorption on AuCu nanoparticles. A DFT Study, Phys. Rev. B., 108, 205414 (2023) (DOI: 10.1103/PhysRevB.108.205414), Q1.

80 I.A. Troyan, D.V. Semenok, A.G. Ivanova, A.V. Sadakov, D. Zhou, **A.G. Kvashnin**, I.A. Kruglov, O.A. Sobolevskiy, M.V. Lyubutina, T. Helm, S.W. Tozer, M. Bykov, A.F. Goncharov, V.M. Pudalov, I.S. Lyubutin, Non-Fermi-Liquid Behavior of Superconducting SnH_4, Adv. Sci., 2303622 (2023) (DOI: 10.1002/advs.202303622), Q1.

79 F. Khorobrykh, S. Klimin, B. Kulnitskiy, F.N. Jalolov, **A.G. Kvashnin**, A. Eliseev, A. Kirichenko, V. Prenas, V. Denisov, N. Mel'nik, P. Sorokin, M. Popov, Cluster structure of ultrahard fullerite revealed by Raman spectroscopy, Carbon, 118314 (2023) (DOI: 10.1016/j.carbon.2023.118314), Q1.

78 A.P. Maltsev, I.V. Chepkasov, **A.G. Kvashnin**, and A.R. Oganov, Ionic Conductivity of Lithium Phosphides, Crystals, 13(5), 756 (2023) (DOI: 10.3390/cryst13050756), Q2

77 S Kaya, H. Dua, U. Sarkar, V.A. Demin, **A.G. Kvashnin**, Reactivity of diamanes against oxidation: A DFT study, Chem. Phys., 571, 111916 (2023) (DOI: 10.1016/j.chemphys.2023.111916), Q2.

76 C. Tantardini, **A.G. Kvashnin**, M. Azizi, X. Gonze, C. Gatti, T. Altalhi, B.I. Yakobson, Electronic Properties of Functionalized Diamanes for Field-Emission Displays, ACS Appl. Mater. Interfaces, 15, 12, 16317–16326 (2023) (DOI: 10.1021/acsami.3c01536), Q1.

75 M. Popov, F. Khorobrykh, S. Klimin, V. Churkin, D. Ovsyannikov, **A. Kvashnin**, Surface Tamm States of 2–5 nm Nanodiamond via Raman Spectroscopy, Nanomat. 13(4), 696 (2023) (DOI: 10.3390/nano13040696), Q1.

74 A.Ya. Pak*, V. Sotskov*, A.A. Gumovskaya, Yu.Z. Vassilyeva, Z.S. Bolatova, Yu.A. Kvashnina, G.Ya. Mamontov, A.V. Shapeev, **A.G. Kvashnin**, Machine learning-Driven Synthesis of $TiZrNbHfTaC_5$ High-Entropy Carbide, npj Comp. Mat., 9, 7 (2023) (DOI: 10.1038/s41524-022-00955-9), Q1, SJR10%.

2022

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2023. “Simulation of multicomponent and high-entropy carbides with subsequent synthesis”, V International Conference and School "Advanced High Entropy Materials", Saint-Petersburg, Russia.

2022. “Digital design of new materials”, XXII International Conference on Science and Technology Russia-Korea-CIS, Moscow, Russia.

2021. “Computational search for new materials for industrial applications”, USPEX Workshops, Skoltech, Moscow, Russia.

2020. “Search for materials with optimal properties”, USPEX Workshops, Skoltech, Moscow, Russia.

2019. “Computational search for supermaterials with optimal properties”, 16th USPEX Workshop, ICTP, Trieste, Italy.

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